HEALTH, SAFETY &ENVIRONMENT

Risk Management

Accident Prevention

The Ministry of Labour and National Service1 postulated six principles of accident prevention in 1956 that are still valid today. These are:

- Accident prevention is an essential part of good management and of good workmanship.
- Management and workers must co-operate wholeheartedly in securing freedom from accidents.
- Top management must take the lead in organising safety in the works.
- There must be a definite and known safety policy in each workplace.
- The organisation and resources necessary to carry out the policy must exist.
- The best available knowledge and methods must be applied.

Role of Risk Management

The role of risk management in industry and commerce is to:

- consider the impact of certain risky events on the performance of the organisation
- devise alternative strategies for controlling these risks and/or their impact on the organisation; and
- relate these alternative strategies to the general decision framework used by the organisation.

Risk Management

- Risk identification may be achieved by a multiplicity of techniques, including:
  - physical inspections
  - management and worker discussions
  - safety audits
  - job safety analysis
  - study of past accidents (can also identify areas of high risk)

- Risk evaluation (or measurement) may be based on:
  - Economic considerations
  - Social considerations
  - Legal considerations

Safe Systems of Work

Safe systems of work are fundamental to accident prevention and should:

- (a) fully document the hazards, precautions and safe working methods
- (b) job training
- (c) be referred to in the 'Arrangements' Section of the Safety Policy.

Safe Systems of Work

The **method study engineers'** aim is to improve methods of production. In this they use a technique known as the SREDIM principle:

- Select (work to be studied);
- Record (how work is done);
- Examine (the total situation);
- Develop (best method for doing work);
- Install (this method into the company's operations);
- Maintain (this defined and measured method).

Safe Systems of Work

- Select the job to be analysed. (SELECT)
- Break the job down into its component parts in an orderly and chronological sequence of job steps. (RECORD)
- Critically observe and examine each component part of the job to determine the risk of accident. (EXAMINE)
- Develop control measures to eliminate or reduce the risk of accident. (DEVELOP)
- Formulate written and safe systems of work and job safety instructions for the job. (INSTALL)
- Review safe systems of work and job safe practices at regular intervals to ensure their utilisation. (MAINTAIN)

Permit-to-Work

A formal, documented safety procedure, forming part of a safe system of work.

Typical applications:

- Hot work (involving naked flames or creation of ignition sources).
- High voltage electrical systems.
- Confined space entry.
- Operational pipelines.
- Excavation near buried services.
- Complex machinery.
- Working at height.

Permit-to-Work

Consists of 4 elements:

1. Issue.
2. Receipt.
3. Clearance/return to service.
4. Cancellation.

Fault Tree Analysis

- Fault tree analysis is a technique that may be utilised to trace back through the chronological progression of causes and effects that have contributed to a particular event, whether it be an accident (industrial safety) or failure (system safety).

- The fault-tree is a logic diagram based on the principle of multicausality that traces all the branches of events that could contribute to an accident or failure.

Fault Tree Analysis

Certain standard symbols are used in the construction of a fault-tree; some of the more common are:

The rectangle identifies an event or contributory factor that results from a combination of contributory factors through the logic gates.

The 'AND' logic gate describes the logical operations whereby the co-existence of *all* input contributory factors are required to produce the output event or contributory factor.

The 'OR' logic gate defines the situation whereby the output event or contributory factor will exist if *any* one of the input contributory factors is present.

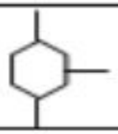
Fault Tree Analysis

Event Symbols in FTA

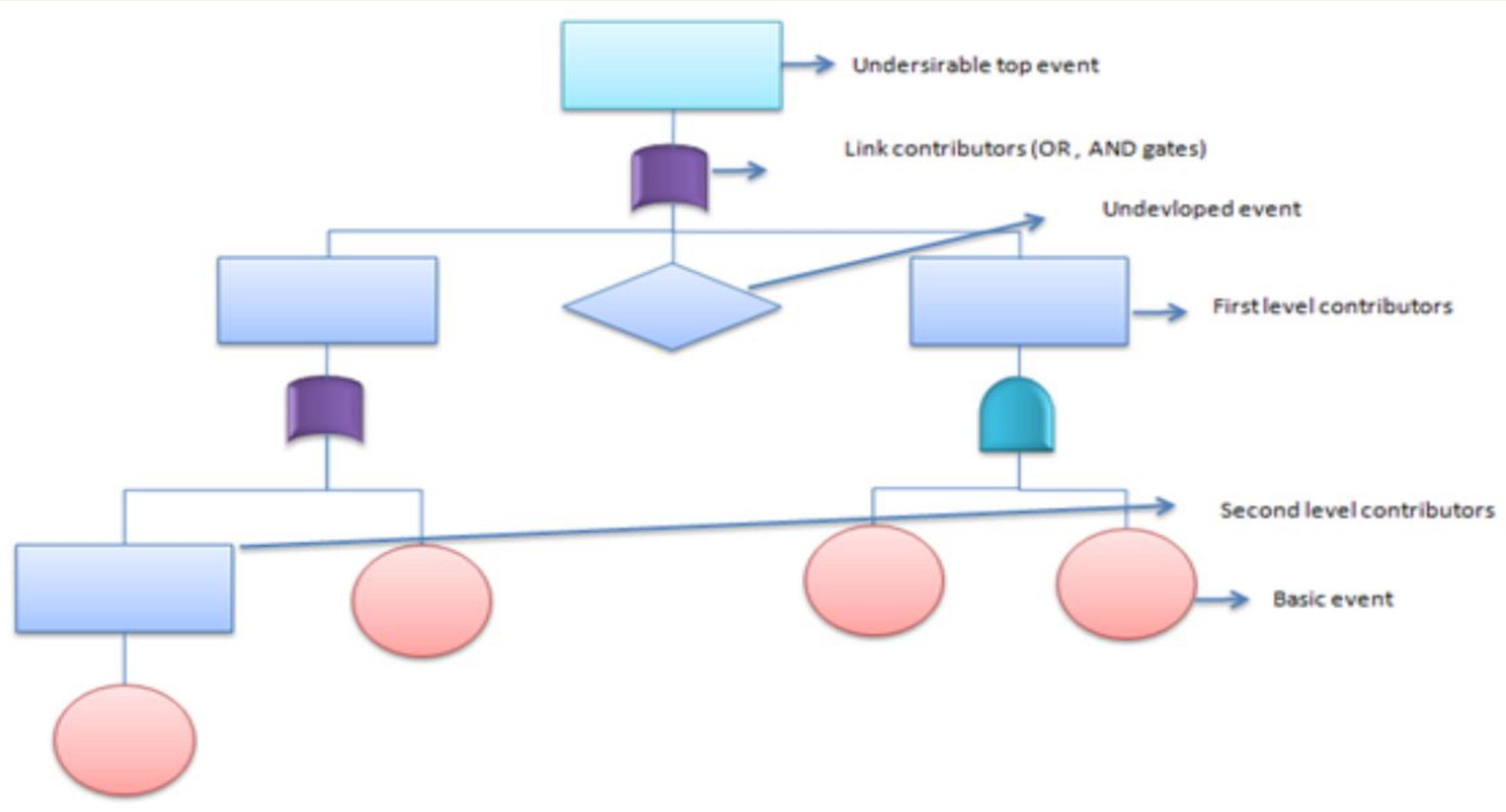
S.No	Event Symbol	Description
1		Primary or basic failure event. It is a random event and sufficient data is available
2		State of system, subsystem or component event
3		Secondary failure or under developed event, can be explored further
4		Conditional event and is associated with the occurrence of some other event
5		House event representing either occurrence or non-occurrence of an event
6		Transfer in and transfer out symbols used to replicate a branch or sub-tree of the FTA

Fault Tree Analysis

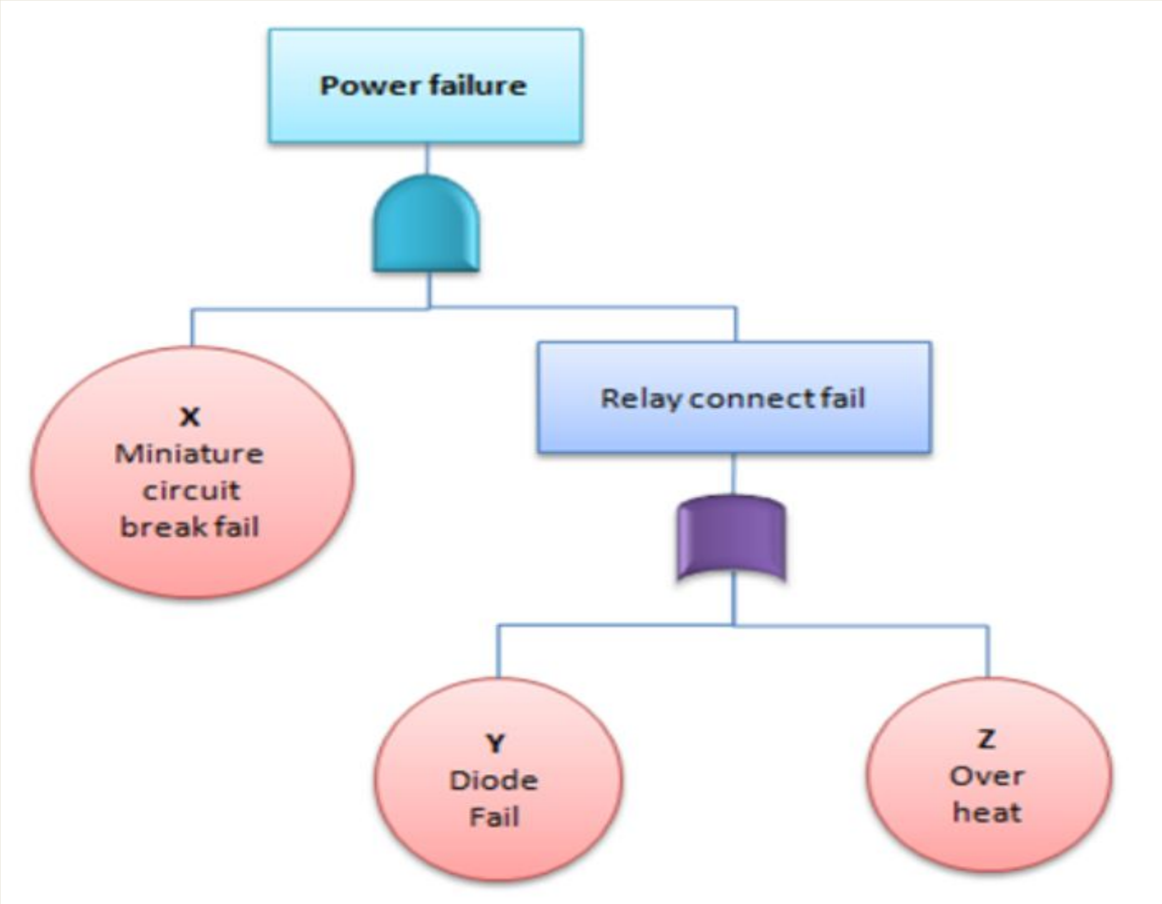
Gate Symbols in FTA

S.No	Gate Symbol	Description
1	 AND Gate	The output event occurs when all the input events occur
2	 OR Gate	The output event occurs when at least one of the input events occur
3	Priority AND Gate 	The output event occurs when all the input events occur in the order from left to right
4	 Exclusive OR gate	The output event occurs if either of the two input events occur but not both
5	 Inhibit gate	The output event occurs when the input event occurs and the attached condition is satisfied

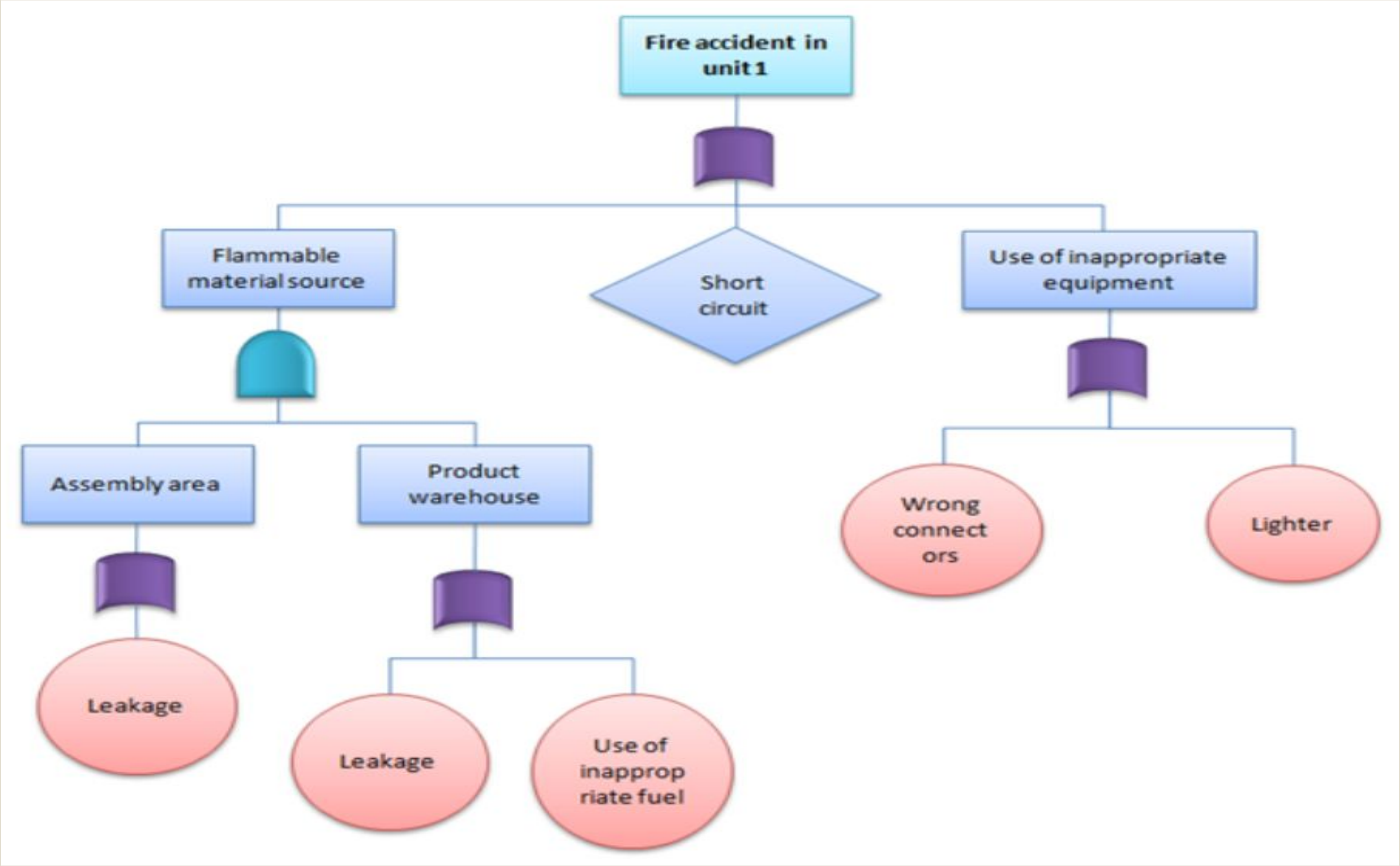
Fault Tree Analysis



Fault Tree Analysis: Example-1



Fault Tree Analysis: Example-2



Cost Benefit Analysis

- Cost benefit analysis techniques have been developed in recent years, as decisions concerning risk management have been made on a cost versus risk basis – i.e. so far as is reasonably practicable.

To undertake a cost benefit analysis, answers to the following questions are required:

- What costs are involved to reduce or eliminate the risk?
- What degree of capital expenditure is required?
- What ongoing costs will be involved, e.g. regular maintenance, training?
- What will the benefits be?
- What is the pay-back period?
- Is there any other more cost-effective method of reducing the risk?

Cost Benefit Analysis

Benefits of effective risk management include:

- few claims resulting in lower insurance premiums,
- less absenteeism,
- fewer injury and damage accidents,
- better levels of health,
- higher productivity/efficiency,
- better utilisation of plant and equipment,
- higher morale and motivation of employees,
- reduction in cost factors.